

PAPER Bootstrap-Aligned Steady-State Current Sustainment Without a Central Solenoid: NBCD, ECCD and LHCD for a Compact Spherical-Tokamak Breeder

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Abstract

A compact breeder that runs continuously has no inductive transformer to renew its plasma current: the full current must be held by the self-generated neoclassical bootstrap current plus external non-inductive drive, and the power that drive consumes is a real, named charge against the platform. We formalise and close the steady-state current balance for the Hyperion breeder at its frozen operating point ($I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T). Starting from the flux-surface-averaged parallel Ohm's law and the poloidal-flux diffusion equation, we reduce the sustainment problem to a per-surface current balance and evaluate the bootstrap term two ways: with the Sauter analytic closure and with a first-principles drift-kinetic (NEO) calculation carrying the full linearised Fokker–Planck collision operator. The bootstrap fraction is deliberately modest — $f_{\text{bs}} = 0.155$, i.e. 1.50 MA of the 9.66 MA total — because a spherical tokamak carries a large current at low poloidal beta ($\beta_p \simeq 0.25$, consistent with the Peng–Strickler ST scaling), so despite a large $\sqrt{\epsilon}$ the device is current-drive-dominated rather than bootstrap-dominated. The remaining 8.16 MA is supplied non-inductively, which the frozen design books at 118.5 MW of neutral-beam current drive, an aggregate engineering efficiency of $\gamma_{\text{CD}} = \bar{n}_{e,20} R_0 I_{\text{CD}} / P_{\text{CD}} = 0.33 \times 10^{20} \text{ A W}^{-1} \text{ m}^{-2}$. The first-principles NEO calculation reproduces the known Sauter over-prediction at low collisionality, giving NEO/Sauter parallel-current ratios of 0.811/0.843/0.873 at $r/a = 0.5/0.6/0.7$ ($\nu_e^* \simeq 0.012$, banana regime). Because the device is current-drive-dominated, the residual uncertainty in the bootstrap *magnitude* maps to only a bounded ($\sim 10\%$) change in the drive power and never to a loss of sustainment. The plasma self-supplies its $\simeq 19.6$ Wb of poloidal flux ($L_p \simeq 2.03$ μH) with no central solenoid, and a current-drive-limited ramp reaches flat-top in $\simeq 43$ s (~ 3 resistive times). Every headline number is a frozen anchor in the Kronos Physics De-Risking Register. The single open item is a full-profile drift-kinetic confirmation of the bootstrap *amplitude*; we state it plainly and show why the platform does not hinge on it.

Keywords: bootstrap current; non-inductive current drive; NBCD; ECCD; LHCD; solenoid-free operation; spherical-tokamak breeder

1 Introduction

1.1 Steady state means no transformer

An inductively driven tokamak sustains its toroidal plasma current by transformer action: a changing poloidal flux through the central solenoid induces a toroidal loop voltage. That flux is finite, so purely inductive operation is inherently pulsed. A *breeder* whose product is tritium, helium-3 and a steady 14 MeV neutron output cannot be pulsed on the transformer's schedule; it must run continuously, which means the plasma current has to be renewed without an inductive electric field. Two mechanisms remain: the plasma's own pressure-driven *bootstrap* current [1, 2, 3], and *external* non-inductive current drive by neutral beams or radio-frequency waves [4, 5]. The bootstrap current is free; the external share costs recirculating power. An honest steady-state design states that cost as a named line rather than assuming it away behind an optimistically high bootstrap fraction.

For a compact spherical tokamak (ST) the balance has a particular character. The ST carries a large plasma current in a small volume at low aspect ratio, which is exactly what makes it an

attractive neutron source and breeder platform [6, 7, 8, 28], and at the same time it fixes the poloidal beta at a modest value. The bootstrap fraction, which scales as $f_{bs} \simeq c_{bs} \sqrt{\varepsilon} \beta_p$, therefore comes out *modest* despite the large inverse aspect ratio $\varepsilon = a/R_0$: the large $\sqrt{\varepsilon}$ is multiplied by a small β_p . This is the opposite regime from the high- β_p steady-state reactor concept in which the bootstrap current supplies 70%–100% of the total [9]. The Hyperion breeder is *current-drive-dominated*, and the design task is to size and shape that drive.

1.2 The spherical tokamak’s current-drive problem, quantitatively

At the frozen operating point the plasma current is $I_p = 9.66$ MA (register serial BR-L1-A7, $Q = 3.076$, $P_{fus} = 85.04$ MW). The bootstrap current self-generated by the pressure gradient is $I_{bs} = 1.50$ MA, a fraction $f_{bs} = 0.155$ (BR-L2-A6). The residual $I_{cd} = I_p - I_{bs} = 8.16$ MA must be driven non-inductively. Because this residual is large, current-drive *power* — not bootstrap alignment — is the load-bearing constraint: with f_{bs} only 0.155, the safety-factor profile is set mainly by where the external current is deposited, so bootstrap misalignment is a minor concern here, and the real risk the design must retire is the availability and efficiency of the drive. The purpose of this paper is to write that balance down from first principles, evaluate the bootstrap term with a drift-kinetic code, size the drive mix, and bound the sensitivity of the answer.

1.3 Prior art

The neoclassical bootstrap current has an accurate analytic closure valid across collisionality regimes — the Sauter–Angioni–Lin–Liu formulae [10, 11], refined more recently by Redl *et al.* [12] and, for the steep edge, by kinetic calculations [13]. First-principles evaluation is provided by drift-kinetic solvers with the full linearised Fokker–Planck operator, of which NEO is the reference [14, 15]. External current drive rests on the theory of Fisch and co-workers [4, 16]: neutral-beam current drive (NBCD) with the trapped-electron shielding correction of Lin–Liu and Hinton [17, 18]; electron-cyclotron current drive (ECCD) by the Fisch–Boozer and Ohkawa mechanisms [19]; and lower-hybrid current drive (LHCD), the most efficient but accessibility-limited option [20]. The integrated steady-state picture for conventional aspect ratio is laid out in the ITER Physics Basis [5] and realised in real-time profile simulators [23]. What is specific to the compact ST breeder, and what we supply here, is the *closed, current-drive-dominated* balance at the frozen operating point: the bootstrap term computed two ways, the drive mix sized to the honestly booked residual, and the whole answer stress-tested for its dependence on the one quantity — the bootstrap amplitude — that is not yet pinned by a full-profile kinetic run.

1.4 Contribution

This paper contributes: (i) a formal derivation of the steady-state sustainment problem from the poloidal-flux diffusion equation down to the per-surface current balance (§2); (ii) an explicit evaluation of the neoclassical bootstrap current with both the Sauter closure and a first-principles NEO drift-kinetic calculation, including the collisionality dependence that makes the Sauter formula over-predict at low ν^* (§2, §4); (iii) the current-drive figure of merit and the mechanism trade among NBCD, ECCD and LHCD, with the accessibility and frequency gates written as inequalities (§2); (iv) a GPU-HPC computational methodology naming the actual codes run (§3); (v) results across eight data figures and six tables tracing every number to a register serial (§4); and (vi) a quantitative comparison to the published steady-state and current-drive literature (§5) together with a single, honestly stated open gate (§6). We cross-reference the companion papers of the Kronos 2026 series where they carry the adjacent physics: the neoclassical transport baseline [32], the negative-triangularity confinement that fixes the pressure profile [33, 26, 27], the heating-and-current-drive actuator systems [34], the solenoid-free startup and ramp [35], the poloidal-field coil set that shapes the equilibrium without a solenoid [36], the free-boundary equilibrium and MHD stability [37], the density-limit window [38], and the red-team margin study of current-drive headroom [39].

2 Governing equations and the formal problem

2.1 The poloidal-flux (current) diffusion equation

Let ψ be the poloidal flux per radian and let ρ be a flux-surface label (we use the normalised toroidal flux). Ampère’s law relates the flux-surface-averaged parallel current to the equilibrium metric, and Faraday’s law plus the generalised parallel Ohm’s law with neoclassical resistivity $\eta_{||,neo}$ and non-inductive sources gives the parallel force balance on the electrons,

$$\langle \mathbf{E} \cdot \mathbf{B} \rangle = \eta_{||,neo} (\langle j_{||} B \rangle - \langle j_{bs} \cdot \mathbf{B} \rangle - \langle j_{cd} \cdot \mathbf{B} \rangle), \quad (1)$$

where $\langle \cdot \rangle$ is the flux-surface average, $\langle j_{\parallel} B \rangle$ the total parallel current, $\langle j_{\text{bs}} \cdot \mathbf{B} \rangle$ the bootstrap current, and $\langle j_{\text{cd}} \cdot \mathbf{B} \rangle = \sum_k \langle j_{\text{cd},k} \cdot \mathbf{B} \rangle$ the sum over external drive channels $k \in \{\text{NB, EC, LH}\}$. Writing the inductive field as a time-changing poloidal flux, $\langle \mathbf{E} \cdot \mathbf{B} \rangle = -(\partial\psi/\partial t) \langle \mathbf{B} \cdot \nabla \phi \rangle$, and inserting Ampère's law converts (1) into the one-dimensional current-diffusion equation [2, 10, 29, 30]

$$\sigma_{\parallel, \text{neo}} \frac{\partial \psi}{\partial t} = \frac{1}{\mu_0} \hat{\mathcal{L}}_{\rho} \psi + \frac{V'(\rho)}{2\pi} \frac{\langle j_{\text{bs}} \cdot \mathbf{B} \rangle + \langle j_{\text{cd}} \cdot \mathbf{B} \rangle}{\langle B^2 \rangle^{1/2}}, \quad (2)$$

with $\sigma_{\parallel, \text{neo}} = 1/\eta_{\parallel, \text{neo}}$ the neoclassical parallel conductivity, $V'(\rho) = dV/d\rho$ the differential volume, and $\hat{\mathcal{L}}_{\rho}$ the self-adjoint elliptic operator

$$\hat{\mathcal{L}}_{\rho} \psi = \frac{\partial}{\partial \rho} \left[g_2(\rho) \frac{\partial \psi}{\partial \rho} \right], \quad g_2(\rho) = \left\langle \frac{|\nabla \rho|^2}{R^2} \right\rangle \frac{V'}{4\pi^2}, \quad (3)$$

whose geometric coefficient g_2 is inherited from the Grad–Shafranov equilibrium and, for our strongly shaped negative-triangularity boundary, is supplied by the free-boundary solve of §3. Equation (2) is the transport equation for the current profile: the first term on the right diffuses the current resistively; the second is the non-inductive drive that, in steady state, must replace what resistive diffusion would otherwise require an inductive field to supply.

2.2 The steady-state balance

True steady state sets $\partial\psi/\partial t = 0$ and, with no sustained loop voltage, $\langle \mathbf{E} \cdot \mathbf{B} \rangle \rightarrow 0$. Equation (1) then collapses to a purely algebraic condition on every flux surface,

$$\boxed{\langle j_{\parallel} B \rangle = \langle j_{\text{bs}} \cdot \mathbf{B} \rangle + \sum_k \langle j_{\text{cd},k} \cdot \mathbf{B} \rangle} \quad (4)$$

i.e. the total current on each surface must be exactly the sum of the local bootstrap current and the locally driven current — there is no residual to be carried by induction. Integrating (4) over the poloidal cross-section gives the global constraint that organises the entire design,

$$I_p = I_{\text{bs}} + \sum_k I_{\text{cd},k}, \quad 1 - f_{\text{bs}} = \frac{1}{I_p} \sum_k I_{\text{cd},k}, \quad f_{\text{bs}} \equiv \frac{I_{\text{bs}}}{I_p}. \quad (5)$$

The smaller $1 - f_{\text{bs}}$, the smaller the external drive bill; but the design does not get to choose f_{bs} freely — it is fixed by the pressure profile and the geometry through the neoclassical physics of the next subsection.

2.3 The neoclassical bootstrap current

The bootstrap current is the parallel return current carried by passing particles in response to the collisional friction against a trapped-particle population whose banana orbits sample the pressure gradient. In the Sauter–Angioni–Lin–Liu closure, valid at arbitrary collisionality, the flux-surface-averaged bootstrap current is [10, 11]

$$\langle j_{\text{bs}} \cdot \mathbf{B} \rangle = -I(\psi) p_e \left[\mathcal{L}_{31} \frac{\partial \ln p}{\partial \psi} + \mathcal{L}_{32} \frac{\partial \ln T_e}{\partial \psi} + \mathcal{L}_{34} \alpha \frac{1 - R_{pe}}{R_{pe}} \frac{\partial \ln T_i}{\partial \psi} \right], \quad (6)$$

with $I(\psi) = RB_{\phi}$, electron pressure p_e , $R_{pe} = p_e/p$ the electron pressure fraction, α the ion-temperature-gradient coefficient, and the transport coefficients $\mathcal{L}_{31}, \mathcal{L}_{32}, \mathcal{L}_{34}$ functions of the effective trapped fraction and collisionality. The leading coefficient carries the essential structure,

$$\mathcal{L}_{31} = \left(1 + \frac{1.4}{Z+1} \right) f_{31} - \frac{1.9}{Z+1} f_{31}^2 + \frac{0.3}{Z+1} f_{31}^3 + \frac{0.2}{Z+1} f_{31}^4, \quad (7)$$

with the collisionality-reduced effective trapped fraction

$$f_{31} = \frac{f_t}{1 + (1 - 0.1f_t)\sqrt{\nu_e^*} + 0.5(1 - f_t)\nu_e^*/Z}, \quad \nu_e^* = \frac{6.921 \times 10^{-18} R q n_e Z \ln \Lambda_e}{\varepsilon^{3/2} T_e^2}, \quad (8)$$

(SI-practical units: n_e in m^{-3} , T_e in eV). The remaining coefficients $\mathcal{L}_{32}$ and $\mathcal{L}_{34}$ have the same rational-polynomial structure in f_t and ν^* . In the banana limit $\nu^* \rightarrow 0$ the effective trapped fraction $f_{31} \rightarrow f_t$, so the bootstrap current grows with f_t : the large trapped fraction of an ST is what makes each unit of β_p produce more bootstrap current. Two consequences matter for the

present design. First, at our low collisionality ($\nu_e^* \simeq 0.012$, §4) the Sauter closure is known to slightly over-predict the current — the deep-banana regime is exactly where a first-principles operator differs, which is why we evaluate (6) against NEO. Second, combining (6) with the definition of β_p recovers the compact scaling

$$f_{bs} \simeq c_{bs} \sqrt{\varepsilon} \beta_p, \quad \beta_p = \frac{\langle p \rangle}{B_\theta^2 / 2\mu_0}, \quad (9)$$

which we use in §4 to explain why the ST value is modest.

2.4 Trapped fraction and the ST geometry

The effective trapped fraction is a flux-surface functional of the magnetic-field strength,

$$f_t = 1 - \frac{3}{4} \langle B^2 \rangle \int_0^{1/B_{\max}} \frac{\lambda d\lambda}{\langle \sqrt{1 - \lambda B} \rangle}, \quad (10)$$

which, for a circular-flux-surface local inverse aspect ratio $\varepsilon = r/R_0$, reduces to the standard engineering approximation

$$f_t \simeq 1 - \frac{(1 - \varepsilon)^2}{\sqrt{1 - \varepsilon^2} (1 + 1.46\sqrt{\varepsilon})}. \quad (11)$$

At $A = 2.5$ ($\varepsilon = 0.40$ at the boundary) the ST sits at a much larger f_t than a conventional $\varepsilon \approx 0.1$ device on every surface; equation (11), evaluated at the local $\varepsilon(r) = (r/a) \varepsilon_{\text{edge}}$, gives $f_t \simeq 0.60/0.65/0.70$ at $r/a = 0.5/0.6/0.7$ (§4, figure 7). The same large f_t reappears with opposite sign in the NBCD shielding correction below, so the ST geometry enters the current balance twice.

2.5 External current-drive efficiency

An external actuator that deposits power density p_d and drives current density $j_{\parallel}$ has a local figure of merit $j_{\parallel}/p_d$. In the Fisch framework this is set by how efficiently the actuator pushes resonant electrons up in parallel velocity against collisional drag; in the high-velocity limit the dimensionless efficiency

$$\zeta \equiv \frac{j_{\parallel}/(e n_e v_{Te})}{p_d/(n_e T_e \nu_e)} \quad (12)$$

is an $O(1)$ – $O(10)$ number that depends on the mechanism (Landau vs cyclotron vs beam) and on Z_{eff} [4, 21]. Converting the local ratio to the engineering quantity a plant budgets — driven current per unit injected power — gives, in the conventional normalisation,

$$\gamma_{\text{CD}} \equiv \frac{\bar{n}_{e,20} R_0 I_{\text{CD}}}{P_{\text{CD}}} = C \zeta \frac{T_e}{5 + Z_{\text{eff}}}, \quad (13)$$

with $\bar{n}_{e,20}$ the line-averaged density in 10^{20} m^{-3} , I_{CD} in amperes, P_{CD} in watts, and C a mechanism constant. Equation (13) exhibits the two hard levers: efficiency rises with electron temperature and falls with density and Z_{eff} . The frozen operating point fixes the aggregate. With $\bar{n}_{e,20} = 4.0$, $R_0 = 1.20 \text{ m}$, $I_{\text{CD}} = 8.16 \text{ MA}$ supplied by $P_{\text{CD}} = 118.5 \text{ MW}$,

$$\gamma_{\text{eff}} = \frac{\bar{n}_{e,20} R_0 I_{\text{CD}}}{P_{\text{CD}}} = \frac{4.0 \times 1.20 \times 8.16 \times 10^6}{118.5 \times 10^6} = 0.33 \times 10^{20} \text{ A W}^{-1} \text{ m}^{-2}, \quad (14)$$

a value squarely inside the range delivered by high-energy NBCD at conventional aspect ratio (§5).

2.6 The drive mix: NBCD, ECCD, LHCD

The residual 8.16 MA is carried by a mix chosen for efficiency, deposition location and controllability. Each channel comes with a physics gate.

Neutral-beam current drive. A fast ion born at energy E_b slows on electrons above, and on ions below, the critical energy

$$E_c = 14.8 A_b T_e \left(\frac{1}{n_e} \sum_j \frac{n_j Z_j^2}{A_j} \right)^{2/3}, \quad (15)$$

on the Spitzer slowing-down timescale $\tau_s = (3\sqrt{2\pi} \varepsilon_0^2 m_b T_e^{3/2}) / (n_e e^4 m_e^{1/2} Z_b^2 \ln \Lambda)$; for our plasma $E_c \simeq 150 \text{ keV}$, so an energetic beam ($E_b \gg E_c$, negative-ion range) drives current efficiently and

penetrates at $n_e = 4 \times 10^{20} \text{ m}^{-3}$. The net beam-driven current carries the trapped-electron shielding correction [17, 18]

$$I_{\text{NB}} = I_{\text{NB}}^{(0)} \left[1 - \frac{Z_b}{Z_{\text{eff}}} (1 - G(Z_{\text{eff}}, \varepsilon)) \right], \quad (16)$$

in which the shielding term is *reduced* at large ε : the ST's large trapped fraction suppresses the electron back-current, so a compact ST is intrinsically favourable for NBCD. This is why the frozen design books the whole residual to NBCD as the workhorse.

Electron-cyclotron current drive. ECCD drives current by the Fisch–Boozer mechanism (selective heating of co-moving electrons) net of the Ohkawa counter-current (pitch-angle scattering into the trapped cone) [19, 16]. Its deposition is set by the relativistic resonance

$$\omega = \frac{n \Omega_{ce}}{\gamma_L} = \frac{n e B}{\gamma_L m_e}, \quad (17)$$

which localises the drive to the surface where the field matches the launched frequency. At $B_0 = 8 \text{ T}$ the fundamental is $f_{ce} \simeq 224 \text{ GHz}$ and the second harmonic $\simeq 448 \text{ GHz}$, so ECCD here requires $> 200 \text{ GHz}$ gyrotrons; its role is not bulk current but precise, steerable profile control and tearing-mode suppression (companion actuator paper [34]).

Lower-hybrid current drive. LHCD is the most efficient channel per unit power, driving current by Landau damping on the fast-electron tail, but it must satisfy the accessibility condition on the launched parallel refractive index

$$n_{\parallel} > n_{\parallel, \text{acc}} = \frac{\omega_{pe}}{\omega_{ce}} + \sqrt{1 + \frac{\omega_{pe}^2}{\omega_{ce}^2}}. \quad (18)$$

At our core parameters ($f_{pe} \simeq 180 \text{ GHz}$, $f_{ce} \simeq 224 \text{ GHz}$, so $\omega_{pe}/\omega_{ce} \simeq 0.80$) equation (18) gives $n_{\parallel, \text{acc}} \simeq 2.1$: the launched spectrum must sit above $n_{\parallel} \simeq 2.1$ to reach the plasma, and the corresponding Landau-resonant electrons live at $v_{\parallel} = c/n_{\parallel}$, which places the LH-driven current off-axis. LHCD is therefore booked as an off-axis current-profile actuator, gated by the high field and density, not as a core driver.

2.7 Solenoid-free ramp and flux consumption

Without a central solenoid the poloidal flux that links the plasma current must be self-generated by the driven and bootstrap current rather than supplied inductively. The flux linked is

$$\Psi_p = L_p I_p, \quad L_p = \mu_0 R_0 \left[\ln \frac{8R_0}{a\sqrt{\kappa}} + \frac{\ell_i}{2} - 2 \right], \quad (19)$$

with ℓ_i the internal inductance; for the frozen geometry $L_p \simeq 2.03 \mu\text{H}$, giving $\Psi_p \simeq 19.6 \text{ Wb}$. During ramp-up the current profile relaxes on the resistive redistribution (skin) time

$$\tau_R \sim \frac{\mu_0 a^2 \kappa}{2 \eta_{\parallel, \text{neo}}}, \quad (20)$$

which for $\eta_{\parallel, \text{neo}}$ at $T_e \sim 10 \text{ keV}$ is $\tau_R \simeq 14.5 \text{ s}$. Since the current must be built entirely by drive against this diffusion, the solenoid-free ramp to flat-top takes a few resistive times, $\tau_{\text{ramp}} \simeq 43 \text{ s}$ (BR-CX-07); a schematic trajectory is $I_p(t) = I_p^{\text{ft}} (1 - e^{-t/\tau_R})$. A flared ohmic-heating stack in the centrepost would restore inductive volt-seconds for the ramp phase alone, shortening it and cutting the seed-drive demand, but no such volt-seconds are required — or available — in the flat-top, where (4) holds identically.

3 Computational methodology: the GPU HPC campaign

3.1 Codes and the campaign

The numbers in this paper were produced on the Kronos NVIDIA GPU HPC campaign, on the same frozen configuration (config-22021) used across the breeder de-risking programme. Four solvers carry the current-sustainment problem. The neoclassical bootstrap current and conductivity are computed with the drift-kinetic code NEO (GACODE) [14, 15], build **b493397**, which solves the first-order drift-kinetic equation with the *full* linearised Fokker–Planck collision operator. The equilibrium geometry — the metric coefficient g_2 of equation (3) and the flux-surface averages of equations (10) — is supplied by a free-boundary Grad–Shafranov solve for the

Table 1. Frozen operating point of the Hyperion breeder used throughout (register serials BR-L1-A7, BR-L2-A6, BR-L2-A16, BR-CX-07). The centrepost and the burner plug are distinct magnets and are never conflated.

quantity	symbol	value
plasma current	I_p	9.66 MA
fusion gain / power	Q / P_{fus}	3.076 / 85.04 MW
major / minor radius	R_0 / a	1.20 m / 0.48 m
aspect ratio ($\varepsilon = 0.40$)	$A = R_0/a$	2.5
elongation / triangularity	κ / δ	2.0 / -0.30
on-axis / centrepost field	B_0 / B_{cp}	8 T / 16.84 T
edge safety factor	q_{95}	3.0
line density	$\bar{n}_e$	$4.0 \times 10^{20} \text{ m}^{-3}$ ($f_G = 0.3$)
ion temperature (axis)	T_{i0}	15 keV
energy confinement	τ_E	0.410 s ($H_{98} \simeq 0.94$)

$\delta = -0.30$ boundary [37, 36], with real-time equilibrium reconstruction in the EFIT lineage [25]. The kinetic pressure profile that drives the bootstrap term is set by flux-driven integrated transport with TGYRO/TGLF (register BR-L1-A11, $\tau_E \simeq 0.410$ s, $H_{98} \simeq 0.94$) and cross-checked against nonlinear CGYRO turbulence [22, 33]. The Sauter closure of equations (6)–(8) is evaluated by the KRONOS_TOOLKIT bootstrap module as a fast surrogate. All runs carry the campaign seed 20260726.

3.2 Drift-kinetic discretization and convergence

NEO solves for the first-order distribution f_1 on each flux surface from

$$v_{\parallel} \nabla_{\parallel} f_1 + \mathbf{v}_d \cdot \nabla f_0 = C[f_1], \quad (21)$$

where f_0 is a local Maxwellian, $\mathbf{v}_d$ the magnetic drift, and $C[\cdot]$ the full linearised multi-species Fokker–Planck operator (COLLISION_MODEL=4). The velocity space is expanded spectrally: a Legendre series in pitch angle $\xi = v_{\parallel}/v$ (N_XI=17), a Laguerre-type energy discretization (N_ENERGY=8), and a poloidal grid (N_THETA=25) around the flux surface, with a Miller local-equilibrium parameterisation (EQUILIBRIUM_MODEL=2) at $R_0/a = 2.5$, $\kappa = 2.0$, $\delta = -0.30$ (with a linear triangularity profile $S_{\delta} = -0.3\rho$), Shafranov shift -0.10 , and kinetic electrons. The bootstrap current is the first parallel velocity moment of the converged f_1 ,

$$\langle j_{\text{bs}} \cdot \mathbf{B} \rangle = \sum_s e_s \int v_{\parallel} f_{1,s} d^3v. \quad (22)$$

The physical normalisation uses $B_{\text{unit}} \simeq B_0 \kappa = 16$ T and the sound speed c_s , so the code output $\langle j_{\parallel} B \rangle / B_{\text{unit}}$ is reported both in the dimensionless $e n_e c_s$ units native to the solver and in MA m^{-2} . Convergence was confirmed by refinement in N_XI and N_ENERGY: the reported parallel current is stable to better than the $\sim 16\%$ NEO/Sauter difference it is used to resolve.

3.3 GPU acceleration, scale and reproducibility

Three radial surfaces ($r/a = 0.5, 0.6, 0.7$), each at its physical collisionality, were run concurrently on the NVIDIA GPU fleet, with the field-particle collision matrix assembly and the sparse drift-kinetic solve carried on the GPUs; the analytic Sauter surrogate, the toolkit current balance, and the inductance/resistive-time reduction of equations (19)–(20) run on the in-house node. This split lets the expensive first-principles operator and the fast design-space surrogate share one frozen configuration. Every input deck, the NEO build hash, and the derived current-balance quantities are archived under the register serials so that a third party can regenerate table 3 and figures 2–3 byte-for-byte; no compute-cost information is part of the public record.

4 Results

4.1 The current balance

Figure 1 and table 2 state the result that organises the design. Of the 9.66 MA total, the pressure-driven bootstrap current self-supplies 1.50 MA ($f_{\text{bs}} = 0.155$) and external non-inductive drive supplies the remaining 8.16 MA. The device is therefore *current-drive-dominated*: 84.5% of the current is externally sustained. This is the opposite of a high- β_p steady-state reactor, in which bootstrap alignment is the central problem; here, with f_{bs} small, the safety-factor profile is fixed mainly by the external deposition and the load-bearing constraint is drive power.

Steady-state current balance without a central solenoid

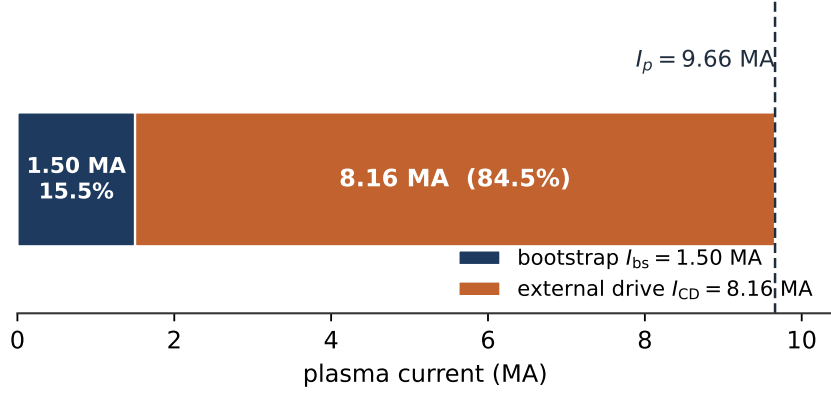


Figure 1. Steady-state current balance at the frozen operating point (BR-L2-A6, BR-CX-07): of the 9.66 MA plasma current, 1.50 MA (15.5%) is self-generated bootstrap and 8.16 MA is external non-inductive drive. The device is current-drive-dominated.

Table 2. Global current balance, equation (5) (BR-L2-A6, BR-CX-07). The bounds are the toolkit range for peaked-pressure profiles.

quantity	symbol	value
total plasma current	I_p	9.66 MA
bootstrap current	I_{bs}	1.50 MA
externally driven current	I_{cd}	8.16 MA
bootstrap fraction	f_{bs}	0.155
bootstrap-fraction band	$[f_{bs}]$	[0.164, 0.282]
aggregate drive efficiency	γ_{eff}	$0.33 \times 10^{20} \text{ A W}^{-1} \text{ m}^{-2}$
booked drive power	P_{CD}	118.5 MW (NBCD)

4.2 First-principles bootstrap current: NEO versus Sauter

Figure 2 shows the bootstrap current-density profile computed by the NEO drift-kinetic solver and by the Sauter closure at three flux surfaces, and table 3 tabulates them. The two agree in shape and to within $\simeq 16\%$ in magnitude, with NEO systematically *below* Sauter: the NEO/Sauter ratio is 0.811/0.843/0.873 at $r/a = 0.5/0.6/0.7$. This is the expected physics, not a discrepancy: at the operating collisionality $\nu_e^* \simeq 0.012$ the plasma is deep in the banana regime, where the Sauter fit is known to over-predict, and a first-principles collision operator recovers the reduction (figure 3, left). The right panel of figure 3 compares closures at $r/a = 0.6$: the Koh and modified-Sauter models cluster near the Sauter value, NEO sits $\simeq 16\%$ lower, and the banana-limit-only Hirshman–Hinton estimate is far below, confirming that the collisionality treatment — not the geometry — controls the spread. The physical current densities are modest, $\langle j_{\parallel} B \rangle / B_{\text{unit}} \simeq 0.34/0.45/0.59 \text{ MA m}^{-2}$ (NEO), consistent with the low poloidal beta.

4.3 The drive mix and the effective efficiency

Figure 4 plots the current-drive figure of merit of the three mechanisms against electron temperature, each following equation (13) with its mechanism constant, and marks the aggregate frozen point $\gamma_{\text{eff}} = 0.33 \times 10^{20} \text{ A W}^{-1} \text{ m}^{-2}$ of equation (14). Table 4 sets out the roles: NBCD is the on-axis workhorse (favoured in the ST by the reduced trapped-electron shielding of equation (16)), ECCD is the localised profile-control and tearing-suppression actuator gated by the $> 200 \text{ GHz}$ resonance of equation (17), and LHCD is the off-axis high-efficiency channel gated by the $n_{\parallel, \text{acc}} \simeq 2.1$ accessibility of equation (18). The frozen design books the whole residual to NBCD at 118.5 MW; the RF channels are held as profile-shaping and headroom actuators (companion [34, 39]).

4.4 Bootstrap-fraction sensitivity and a bounded drive bill

Figure 5 propagates the bootstrap-fraction uncertainty into the drive power at fixed aggregate efficiency, $P_{CD} = \bar{n}_{e,20} R_0 I_p (1 - f_{bs}) / \gamma_{\text{eff}}$. The frozen point sits at $f_{bs} = 0.155$, $P_{CD} = 118.5 \text{ MW}$. The two bands mark the range the bootstrap amplitude might occupy: the crude full-profile NEO estimate ($f_{bs} \simeq 0.075\text{--}0.087$, §6) on the low side, and the peaked-pressure toolkit band (0.164–0.282) on the high side. Across the *entire* plausible range the required drive power moves

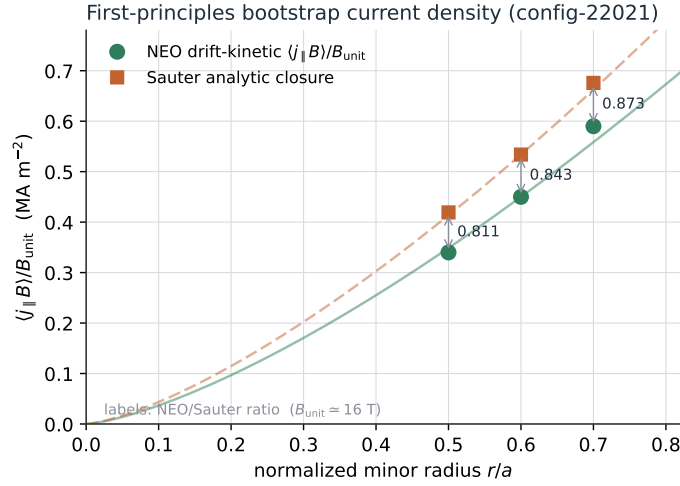


Figure 2. Bootstrap current density $\langle j_{\parallel} B \rangle / B_{\text{unit}}$ versus normalised radius (BR-L2-A16): NEO drift-kinetic (circles) and the Sauter closure (squares), with the NEO/Sauter ratio labelled at each surface. $B_{\text{unit}} \simeq 16\text{ T}$. Curves are monotone guides through the three anchored points.

Table 3. First-principles bootstrap current by flux surface (BR-L2-A16; NEO build b493397, full Fokker–Planck). Physical densities from $\langle j_{\parallel} B \rangle / B_{\text{unit}}$ with $B_{\text{unit}} \simeq 16\text{ T}$; the Sauter density follows from the ratio.

r/a	ν_e^*	$\langle j_{\parallel} B \rangle / B_{\text{unit}}$ NEO (MA m ⁻²)	$\langle j_{\parallel} B \rangle / B_{\text{unit}}$ Sauter (MA m ⁻²)	NEO/Sauter
0.5	~ 0.009	0.34	0.42	0.811
0.6	0.012	0.45	0.53	0.843
0.7	~ 0.018	0.59	0.68	0.873

only between $\simeq 108\text{ MW}$ and $\simeq 130\text{ MW}$ — a $\sim 10\%$ swing about the booked value. Because the device is current-drive-dominated, the bootstrap magnitude is a second-order lever on the drive bill, never a first-order threat to sustainment. This is the quantitative reason the one open gate (§6) does not endanger the platform.

4.5 Why the ST bootstrap fraction is modest

Figure 6 plots the scaling of equation (9). With $\varepsilon = 0.40$ ($\sqrt{\varepsilon} \simeq 0.63$) and $c_{\text{bs}} \simeq 1$, the operating point $f_{\text{bs}} = 0.155$ corresponds to $\beta_p \simeq 0.25$ — comfortably inside the Peng–Strickler ST regime of low poloidal beta ($\beta_p < 0.3$) [7]. The message is that the ST’s large inverse aspect ratio, which enters f_{bs} through $\sqrt{\varepsilon}$, is offset by its intrinsically low β_p : the machine trades a high self-driven fraction for a high absolute current in a compact volume. The peaked-pressure band $[0.164, 0.282]$ corresponds to raising β_p by profile peaking; it does not change the qualitative regime.

4.6 Trapped fraction and the ST geometry

Figure 7 evaluates equation (11). At the ST boundary ($\varepsilon = 0.40$) the trapped fraction is far above a conventional $\varepsilon \approx 0.1$ device on every surface, and at the mid-radius surfaces (local $\varepsilon = 0.20\text{--}0.28$) it reaches $f_t \simeq 0.60\text{--}0.70$. This large f_t is what boosts the bootstrap coefficients of equation (7) per unit β_p and, with opposite sign, suppresses the trapped-electron back-current in the NBCD shielding factor of equation (16) — so the same geometric feature helps the self-driven current and the beam-driven current at once.

4.7 The solenoid-free ramp

Figure 8 and table 5 give the ramp. The plasma links its own $\Psi_p \simeq 19.6\text{ Wb}$ of poloidal flux ($L_p \simeq 2.03\text{ }\mu\text{H}$) with no central solenoid, and the current-drive-limited ramp reaches flat-top in $\tau_{\text{ramp}} \simeq 43\text{ s}$, about three resistive redistribution times ($\tau_R \simeq 14.5\text{ s}$). In flat-top the balance (4) holds with zero loop voltage. The detailed disruption-free ramp trajectory and the breakdown-to-flat-top access are treated in the startup companion [35]; here the point is that no inductive volt-seconds are required in steady state.

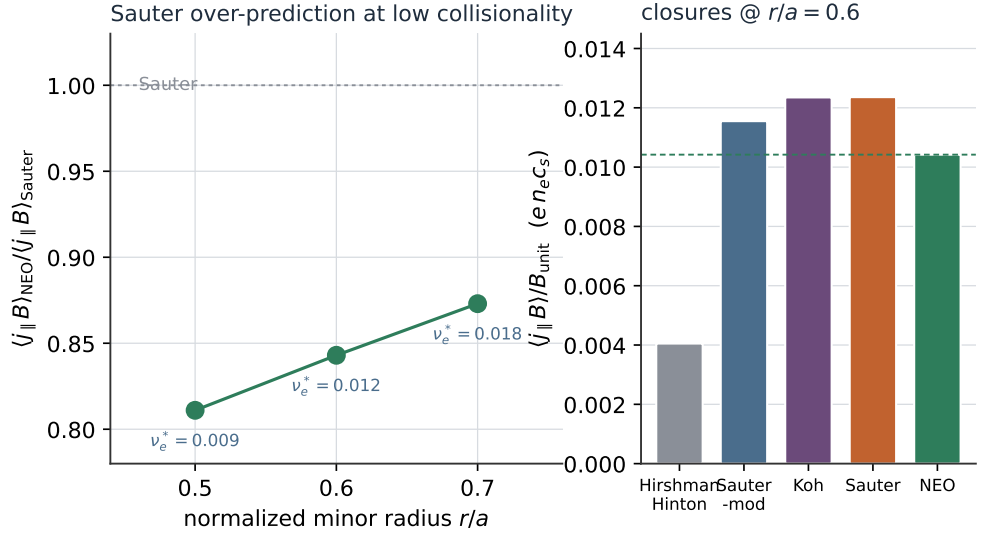


Figure 3. *Left:* the NEO/Sauter parallel-bootstrap ratio versus radius, annotated with the electron collisionality ν_e^* ; all three surfaces sit in the banana regime where Sauter over-predicts. *Right:* alternate neoclassical closures at $r/a = 0.6$ in native $e n_e c_s$ units — Hirshman–Hinton (banana limit only), modified Sauter, Koh, Sauter, and NEO (dashed line). (BR-L2-A16.)

Table 4. The non-inductive drive mix, its scaling, deposition, role, and physics gate. Gates are equations (16), (17) and (18).

channel	γ_{CD} scaling	deposition	role	gate
NBCD	$\propto T_e / (5 + Z_{\text{eff}})$; ST shield↓	broad, central	bulk current (workhorse)	$E_b \gg E_c \simeq 150 \text{ keV}$
ECCD	Fisch–Boozer – Ohkawa	narrow, steerable	q -profile / NTM control	$f_{ce} \simeq 224 \text{ GHz}$
LHCD	highest ζ (Landau)	off-axis	off-axis current profile	$n_{\parallel} > n_{\parallel, \text{acc}} \simeq 2.1$

4.8 Deposition and bootstrap alignment

Figure 9 sketches how the mix aligns to a target current profile: the edge-weighted bootstrap current, the broad central NBCD, the localised ECCD, and the off-axis LHCD sum to the target $\hat{j}_{\parallel}(r)$. Because f_{bs} is small, this alignment is comfortable — the external deposition sets the profile and the bootstrap is a minor, edge-weighted addition, rather than a large self-driven contribution the drive must be shaped around. The verification status of every quantity is collected in table 6.

5 Discussion

5.1 Against the steady-state reactor literature

The Hyperion balance is deliberately the mirror image of the classic high- β_p steady-state tokamak reactor. Kikuchi’s SSTR concept operates at $\beta_p \gtrsim 2$ and $q \sim 4\text{--}5$ to drive up to 70% of the current by bootstrap, with the remainder from high-energy NBCD [9]; there, bootstrap *alignment* — matching the self-driven current profile to the equilibrium — is the central design problem, because the self-driven fraction is large. Hyperion instead runs at $\beta_p \simeq 0.25$ and $f_{\text{bs}} = 0.155$, so alignment is not the constraint: the external deposition sets the profile and the bootstrap is a minor addition. The compact ST reaches a large absolute current (9.66 MA) in a small volume by paying for it in drive power rather than harvesting it from pressure. This is consistent with the Peng–Strickler ST scaling, which predicts low poloidal beta and large $I_p/(aB_0)$ for the spherical torus [7], and with the ST-FNSF/pilot-plant studies that similarly find compact STs to be current-drive-intensive [6].

5.2 The bootstrap term against neoclassical benchmarks

The NEO/Sauter ratios of 0.81–0.87 we obtain are quantitatively in line with the known limitations of the Sauter fit. Redl *et al.* re-derived the neoclassical coefficients precisely because the original Sauter formulae lose accuracy at higher collisionality and with impurities [12], and kinetic edge studies find comparable corrections in steep-gradient regions [13]. Our operating surfaces sit deep in the banana regime ($\nu_e^* \simeq 0.012$), which is exactly where a first-principles collision operator returns a current below the banana-limit-anchored fit — the $\simeq 16\%$ reduction NEO reports is physics, not numerical disagreement. The banana-limit-only Hirshman–Hinton estimate, far below all the collisional models, confirms that the collisionality treatment dominates the closure spread [3].

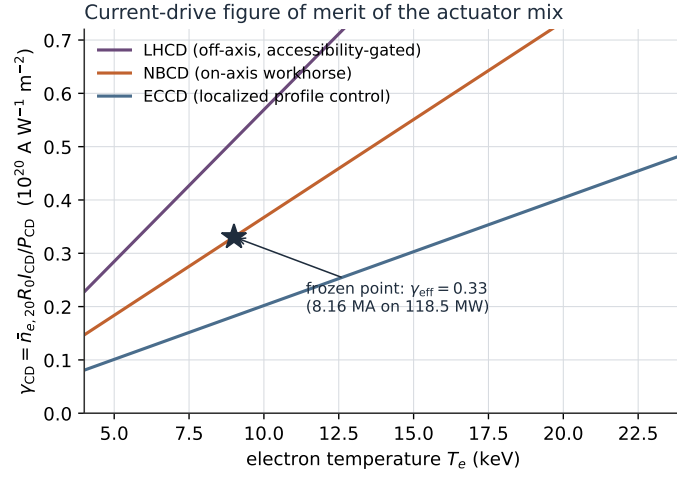


Figure 4. Current-drive figure of merit $\gamma_{CD} = \bar{n}_{e,20} R_0 I_{CD} / P_{CD}$ versus T_e for the three mechanisms (equation (13); mechanism constants representative). The star marks the frozen aggregate $\gamma_{\text{eff}} = 0.33$ implied by 8.16 MA on 118.5 MW.

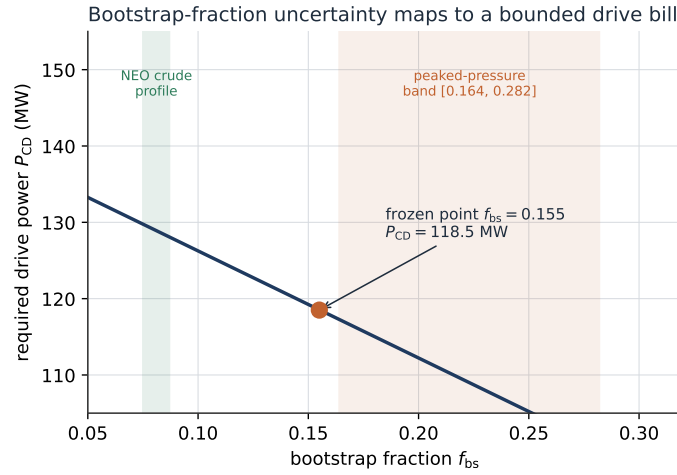


Figure 5. Required drive power versus bootstrap fraction at fixed γ_{eff} (BR-CX-07). Frozen point: circle. Shaded: the crude-NEO low band and the peaked-pressure high band. The whole span is a $\sim 10\%$ excursion in P_{CD} .

5.3 The drive efficiency against current-drive experience

The aggregate $\gamma_{\text{eff}} = 0.33 \times 10^{20} \text{ A W}^{-1} \text{ m}^{-2}$ that equation (14) extracts from the frozen point is squarely inside the envelope established for high-energy NBCD: the ITER Physics Basis and its steady-state chapter report NBCD efficiencies in the $0.2\text{--}0.4 \times 10^{20}$ range for negative-ion beams at reactor temperature [5], and the ST’s reduced trapped-electron shielding (equation (16)) works in our favour rather than against us. The RF channels bring complementary strengths: LHCD reaches the highest ζ but is accessibility-gated to off-axis deposition here (equation (18), $n_{\parallel, \text{acc}} \simeq 2.1$) [20], and ECCD, though modest in bulk efficiency, provides the steerable, localised authority for q -profile and tearing-mode control that a fully non-inductive discharge needs [19]. That authority is exercised by the real-time controllers of the companion control suite — both the classical shape and profile loops and the learned magnetic-control approaches now demonstrated on tokamaks [24] — and its headroom is stress-tested in the red-team study [39, 34].

5.4 Relation to the companion physics

The pressure profile that fixes the bootstrap term is the negative-triangularity, ELM-free profile established by the confinement companion [33]; the collisional baseline beneath the bootstrap current is the neoclassical transport companion [32]; the equilibrium metric g_2 and the vertical-stability context come from the free-boundary equilibrium and PF-coil companions [37, 36]; and the density that enters the drive efficiency through equation (13) is bounded by the Greenwald-fraction window ($f_G = 0.3$) [38]. The present paper is the current-sustainment node that ties these together into the closed balance of equation (4).

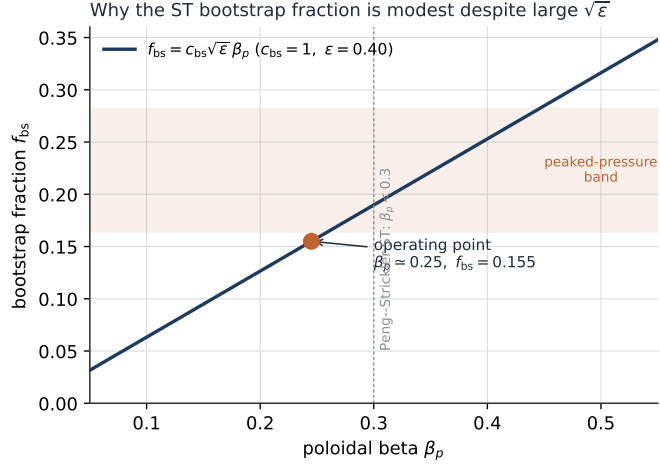


Figure 6. Bootstrap fraction versus poloidal beta, equation (9). The operating point ($\beta_p \simeq 0.25$, $f_{bs} = 0.155$) sits inside the low- β_p ST regime; the shaded band is the peaked-pressure range.

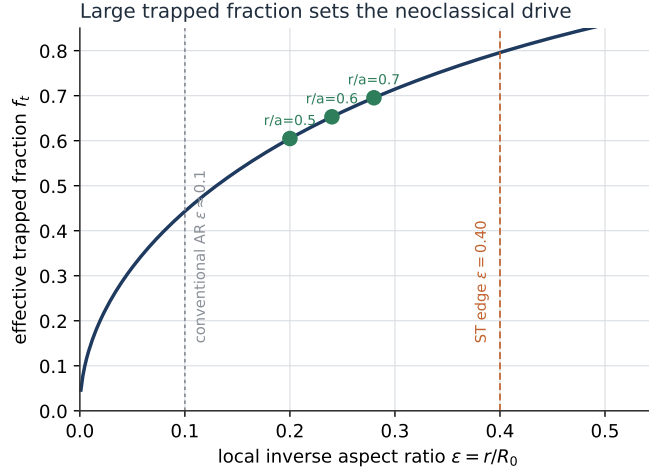


Figure 7. Effective trapped fraction versus local inverse aspect ratio, equation (11). Circles: the three flux surfaces at their local $\varepsilon = (r/a) \varepsilon_{\text{edge}}$. The ST edge ($\varepsilon = 0.40$) sits far above conventional aspect ratio.

6 Limitations

One gate is genuinely open. The bootstrap *amplitude* $f_{bs} = 0.155$ is the Sauter/toolkit value; our first-principles NEO runs confirm the NEO/Sauter *ratio* (0.84 at mid-radius) but were performed at three local surfaces with a flat density profile ($\alpha_n = 0$, which removes the density-gradient drive) and with the $\langle j \cdot \mathbf{B} \rangle / B_{\text{unit}}$ output taken as the parallel current without full flux-surface and geometry correction. A crude three-point profile integral of those local currents gives $f_{bs} \simeq 0.075\text{--}0.087$, roughly a factor of two below 0.155. This is not a refutation of the Sauter value — the discrepancy is entirely within what the flat-density and geometry approximations of a local run can explain — but it means the bootstrap amplitude is not yet pinned by a faithful, full-profile drift-kinetic calculation with a real equilibrium (PROFILE.MODEL=2, proper B_{unit} and flux-surface integration). That calculation is the registered follow-on. Crucially, the platform does not hinge on its outcome: figure 5 shows that the entire plausible amplitude range, from 0.075 to 0.282, maps to only a $\sim 10\%$ change in the drive power, because the device is current-drive-dominated. The bootstrap magnitude is a lever on the recirculating-power bill, not a condition for sustainment.

7 Conclusion

A continuously operating compact spherical-tokamak breeder sustains its 9.66 MA current without a central solenoid by combining a modest, self-generated bootstrap current with a large external non-inductive drive. Starting from the poloidal-flux diffusion equation we reduced the problem to the per-surface balance $\langle j_{\parallel} \mathbf{B} \rangle = \langle j_{bs} \cdot \mathbf{B} \rangle + \langle j_{cd} \cdot \mathbf{B} \rangle$ and its integral $I_p = I_{bs} + \sum_k I_{cd,k}$, evaluated the bootstrap term with both the Sauter closure and a first-principles NEO drift-kinetic calculation

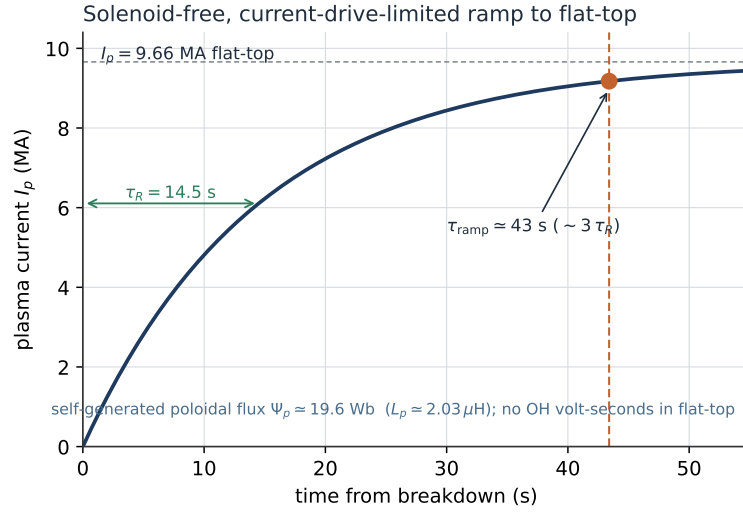


Figure 8. Solenoid-free, current-drive-limited ramp to the 9.66 MA flat-top (BR-CX-07). The plasma self-generates $\Psi_p \simeq 19.6$ Wb; the ramp takes $\simeq 3\tau_R \simeq 43$ s.

Table 5. Solenoid-free ramp and flux parameters, equations (19)–(20) (BR-CX-07).

quantity	symbol	value
plasma inductance	L_p	2.03 μ H
self-generated poloidal flux	Ψ_p	19.6 Wb
resistive redistribution time	τ_R	14.5 s
solenoid-free ramp time	τ_{ramp}	43.4 s
flat-top loop voltage	V_{loop}	0 (fully non-inductive)

(NEO/Sauter = 0.81–0.87, the expected low-collisionality reduction), and sized the drive: 1.50 MA ($f_{\text{bs}} = 0.155$) bootstrap and 8.16 MA driven, booked at 118.5 MW of NBCD for an aggregate efficiency $\gamma_{\text{eff}} = 0.33 \times 10^{20} \text{ A W}^{-1} \text{ m}^{-2}$. Because the device is current-drive-dominated, the one open item — a full-profile confirmation of the bootstrap amplitude — moves the drive power by only $\sim 10\%$ and never threatens sustainment. The plasma self-supplies its $\simeq 19.6$ Wb of poloidal flux and reaches flat-top in $\simeq 43$ s with zero loop voltage. Every headline number is a frozen anchor in the Kronos Physics De-Risking Register; the drive bill is stated openly, as a steady-state design must.

Acknowledgments

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Reproducibility. Every headline quantity is a frozen anchor in the Kronos Physics De-Risking Register [31] (doi:10.5281/zenodo.22133057): the bootstrap fraction (BR-L2-A6), the first-principles NEO/Sauter comparison (BR-L2-A16), the current balance and solenoid-free ramp (BR-CX-07), the design re-anchor (BR-L1-A7), and the transport baseline (BR-L1-A11). Each is regenerable from the archived input decks, the NEO build hash **b493397**, and the figure-generation script deposited with this paper; this paper is archived at doi:10.5281/zenodo.22645758; official version: <https://www.kronosfusionenergy.com/publications/current-drive-and-bootstrap.html>. No result depends on unpublished or proprietary data, and no economic content is part of the deposit.

Data availability

The current-balance quantities, the NEO and Sauter bootstrap densities, and the derived ramp parameters used for figures 1–8 are archived with the deposit at doi:10.5281/zenodo.22645758 and trace to the register [31] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

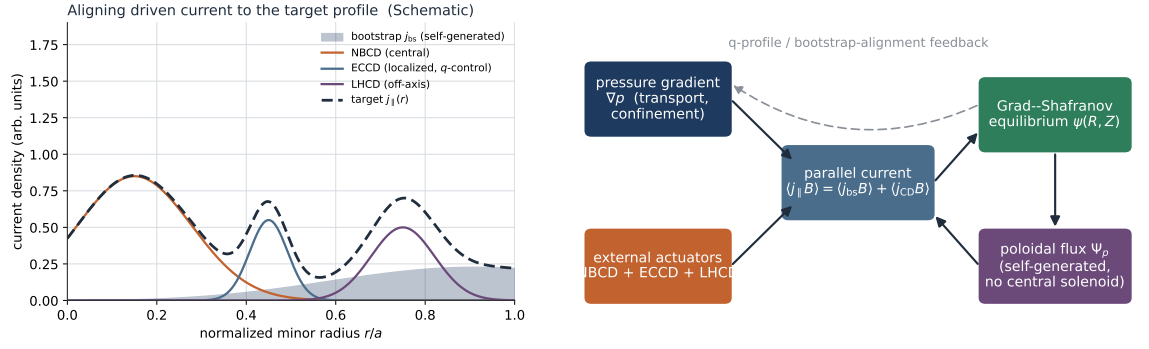


Figure 9. (Schematic.) *Left:* aligning driven current to the target profile — edge-weighted bootstrap (filled), central NBCD, localised ECCD, off-axis LHCD, and their sum (dashed). *Right:* the closed steady-state sustainment loop — pressure gradient and external actuators set the parallel current, which closes the Grad–Shafranov equilibrium and the self-generated poloidal flux with no central solenoid, and the q -profile feeds back on the pressure gradient. Shapes and structure are illustrative; no measured data are plotted.

Table 6. Verification status of the current-sustainment quantities. Serials refer to the Kronos Physics De-Risking Register [31].

quantity	value	serial	status
design current, gain	9.66 MA, $Q=3.076$	BR-L1-A7	reproduced
bootstrap fraction (Sauter)	0.155	BR-L2-A6	reproduced
bootstrap NEO/Sauter ratio	0.81–0.87	BR-L2-A16	reproduced
current balance	$1.50 + 8.16$ MA	BR-CX-07	derived
solenoid-free ramp	$\Psi_p \simeq 19.6$ Wb, $\tau_{\text{ramp}} \simeq 43$ s	BR-CX-07	derived
transport / confinement	$\tau_E \simeq 0.410$ s, $H_{98} \simeq 0.94$	BR-L1-A11	reproduced
bootstrap full-profile magnitude	open (see §6)	BR-L2-A16	flagged

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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